SOLUTIONS TO END-OF-CHAPTER PROBLEMS

7-1 $D_0 = \$1.50$; $g_{1-3} = 5\%$; $g_n = 10\%$; D_1 through D_5 = ?

$D_1 = D_0(1 + g_1) = \$1.50(1.05) = \$1.5750.$
$D_2 = D_0(1 + g_1)(1 + g_2) = \$1.50(1.05)^2 = \$1.6538.$
$D_3 = D_0(1 + g_1)(1 + g_2)(1 + g_3) = \$1.50(1.05)^3 = \$1.7364.$
$D_4 = D_0(1 + g_1)(1 + g_2)(1 + g_3)(1 + g_n) = \$1.50(1.05)^3(1.10) = \$1.9101.$
$D_5 = D_0(1 + g_1)(1 + g_2)(1 + g_3)(1 + g_n)^2 = \$1.50(1.05)^3(1.10)^2 = \$2.1011.$

7-2 $D_1 = \$1.50$; $g = 7\%$; $r_s = 15\%$; $\hat{P}_0$ = ?

$$\hat{P}_0 = \frac{D_1}{r_s - g} = \frac{\$1.50}{0.15 - 0.07} = \$18.75.$$

7-3 $P_0 = \$20$; $D_0 = \$1.00$; $g = 10\%$; $\hat{P}_1$ = ?; $\hat{r}_s$ = ?

$$\hat{P}_1 = P_0(1 + g) = \$20(1.10) = \$22.$$

$$\hat{r}_s = \frac{D_1}{P_0} + g = \frac{\$1.00(1.10)}{\$20} + 0.10$$

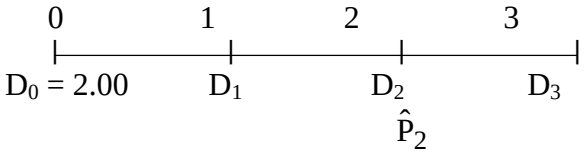
$$= \frac{\$1.10}{\$20} + 0.10 = 15.50\%. \quad \hat{r}_s = 15.50\%.$$

7-4 $D_{ps} = \$5.00$; $V_{ps} = \$50$; r_{ps} = ?

$$r_{ps} = \frac{D_{ps}}{v_{ps}} = \frac{\$5.00}{\$50.00} = 10\%.$$

© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

7-5



Step 1: Calculate the required rate of return on the stock:

$$r_s = r_{RF} + (r_M - r_{RF})b = 7.5\% + (4\%)1.2 = 12.3\%.$$

Step 2: Calculate the expected dividends:

$D_0 = \$2.00$
$D_1 = \$2.00(1.20) = \$2.40$
$D_2 = \$2.00(1.20)^2 = \$2.88$
$D_3 = \$2.88(1.07) = \$3.08$

Step 3: Calculate the PV of the expected dividends:

$$PV_{Div} = \$2.40/(1.123) + \$2.88/(1.123)^2 = \$2.14 + \$2.28 = \$4.42.$$

Step 4: Calculate $\hat{P}_2$:

$$\hat{P}_2 = D_3/(r_s - g) = \$3.08/(0.123 - 0.07) = \$58.11.$$

Step 5: Calculate the PV of $\hat{P}_2$:

$$PV = \$58.11/(1.123)^2 = \$46.08.$$

Step 6: Sum the PVs to obtain the stock's price:

$$\hat{P}_0 = \$4.42 + \$46.08 = \$50.50.$$

Alternatively, using a financial calculator, input the following:

$CF_0 = 0$, $CF_1 = 2.40$, and $CF_2 = 60.99$ (2.88 + 58.11) and then enter I/YR = 12.3 to solve for NPV = \$50.50.

© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

7-6 The problem asks you to determine the constant growth rate, given the following facts: $P_0 = \$80$, $D_1 = \$4$, and $r_s = 14\%$. Use the constant growth rate formula to calculate g:

$$\hat{r}_s = \frac{D_1}{P_0} + g$$

$$0.14 = \frac{\$4}{\$80} + g$$

$$g = 0.09 = 9\%.$$

© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

7-7 The problem asks you to determine the value of $\hat{P}_3$, given the following facts: $D_1 = \$2$, $b = 0.9$, $r_{RF} = 5.6\%$, $RP_M = 6\%$, and $P_0 = \$25$. Proceed as follows:

Step 1: Calculate the required rate of return:

$$r_s = r_{RF} + (r_M - r_{RF})b = 5.6\% + (6\%)0.9 = 11\%.$$

Step 2: Use the constant growth rate formula to calculate g:

$$\hat{r}_s = \frac{D_1}{P_0} + g$$
$$0.11 = \frac{\$2}{\$25} + g$$
$$g = 0.03 = 3\%.$$

Step 3: Calculate $\hat{P}_3$:

$$\hat{P}_3 = P_0(1 + g)^3 = \$25(1.03)^3 = \$27.3182 \approx \$27.32.$$

Alternatively, you could calculate D_4 and then use the constant growth rate formula to solve for $\hat{P}_3$:

$D_4 = D_1(1 + g)^3 = \$2.00(1.03)^3 = \$2.1855.$
$\hat{P}_3 = \$2.1855/(0.11 - 0.03) = \$27.3188 \approx \$27.32.$

7-8 $V_{ps} = D_{ps}/r_{ps}$; therefore, $r_{ps} = D_{ps}/V_{ps}$.

a. $r_{ps} = \$8/\$60 = 13.3\%$.
b. $r_{ps} = \$8/\$80 = 10\%$.
c. $r_{ps} = \$8/\$100 = 8\%$.
d. $r_{ps} = \$8/\$140 = 5.7\%$.

© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

7-9 $$\hat{P}_0 = \frac{D_1}{r_s - g} = \frac{D_0(1+g)}{r_s - g} = \frac{\$5[1+(-0.04)]}{0.15-(-0.04)]} = \frac{\$4.80}{0.19} = \$25.26.$$

7-10 a. $r_i = r_{RF} + (r_M - r_{RF})b_i$.
$r_C = 9\% + (13\% - 9\%)0.4 = 10.6\%$. $r_D = 9\% + (13\% - 9\%)\text{-}0.5 = 7\%$.

Note that r_D is *below* the risk-free rate. But since this stock is like an insurance policy because it "pays off" when something bad happens (the market falls), the low return is not unreasonable.

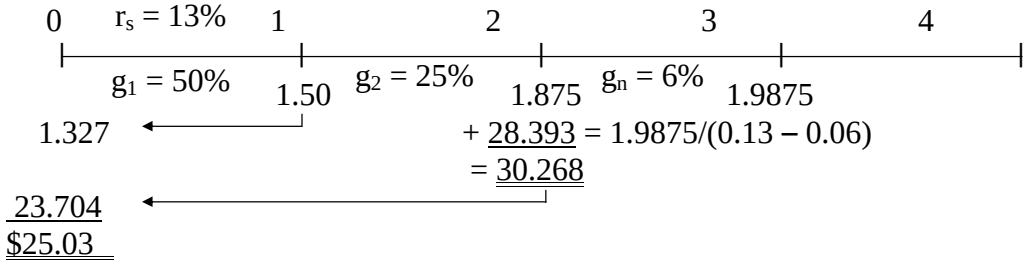
b. In this situation, the expected rate of return is as follows:
$\hat{r}_c = D_1/P_0 + g = \$1.50/\$25 + 4\% = 10\%$.

However, the required rate of return is 10.6%. Investors will seek to sell the stock, dropping its price to the following:

$$\hat{P}_C = \frac{\$1.50}{0.106 - 0.04} = \$22.73.$$

At this point, $\hat{r}_c = \frac{\$1.50}{\$22.73} + 4\% = 10.6\%$, and the stock will be in equilibrium.

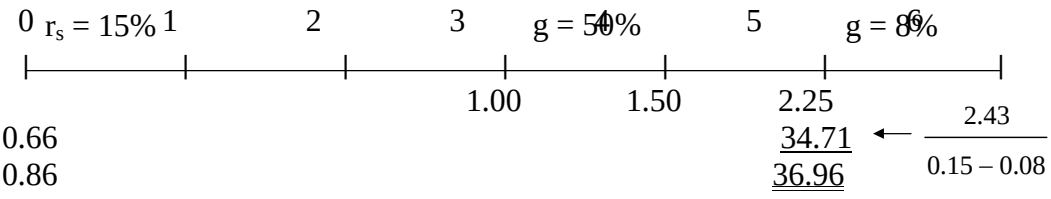
7-11 $D_0 = \$1$, $r_S = 7\% + 6\% = 13\%$, $g_1 = 50\%$, $g_2 = 25\%$, $g_n = 6\%$.



© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

7-12 Calculate the dividend stream and place them on a time line. Also, calculate the price of the stock at the end of the supernormal growth period, and include it, along with the dividend to be paid at t = 5, as CF_5. Then, enter the cash flows as shown on the time line into the cash flow register, enter the required rate of return as I = 15, and then find the value of the stock using the NPV calculation. Be sure to enter $CF_0 = 0$, or else your answer will be incorrect.

$D_0 = 0$; $D_1 = 0$, $D_2 = 0$, $D_3 = 1.00$
$D_4 = 1.00(1.5) = 1.5$; $D_5 = 1.00(1.5)^2 = 2.25$; $D_6 = 1.00(1.5)^2(1.08)$ = \$2.43.
$\hat{P}_0$ = ?



0.66
0.86
18.38
34.71
36.96
\$19.89 = $\hat{P}_0$

$\hat{P}_5 = D_6/(r_s - g) = 2.43/(0.15 - 0.08) = 34.71$. This is the price of the stock at the end of Year 5.

$CF_0 = 0$; $CF_{1\text{-}2} = 0$; $CF_3 = 1.0$; $CF_4 = 1.5$; $CF_5 = 36.96$; I = 15%.

With these cash flows in the CFLO register, press NPV to get the value of the stock today: NPV = \$19.89.

7-13 a. $V_{ps} = \dfrac{D_{ps}}{r_{ps}} = \dfrac{\$10}{0.08} = \$125.$

b. $V_{ps} = \dfrac{\$10}{0.12} = \$83.33.$

© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

7-14 a. $g = \$1.1449/\$1.07 - 1.0 = 7\%$.

Calculator solution: Input N = 1, PV = -1.07, PMT = 0, FV = 1.1449, I/YR = ? I = 7.00%.

b. $\$1.07/\$21.40 = 5\%$.

c. $\hat{r}_s = D_1/P_0 + g = \$1.07/\$21.40 + 7\% = 5\% + 7\% = 12\%$.

7-15 a. 1. $\hat{P}_0 = \dfrac{\$2(1-0.05)}{0.15+0.05} = \dfrac{\$1.90}{0.20} = \$9.50$.

2. $\hat{P}_0 = \$2/0.15 = \$13.33$.

3. $\hat{P}_0 = \dfrac{\$2(1.05)}{0.15-0.05} = \dfrac{\$2.10}{0.10} = \$21.00$.

4. $\hat{P}_0 = \dfrac{\$2(1.10)}{0.15-0.10} = \dfrac{\$2.20}{0.05} = \$44.00$.

b. 1. $\hat{P}_0 = \$2.30/0 =$ Undefined.

2. $\hat{P}_0 = \$2.40/(-0.05) = -\$48$, which is nonsense.

These results show that the formula does not make sense if the required rate of return is equal to or less than the expected growth rate.

c. No.

© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

7-16 a. $r_s = r_{RF} + (r_M - r_{RF})b = 11\% + (14\% - 11\%)1.5 = 15.5\%$.
$\hat{P}_0 = D_1/(r_s - g) = \$2.25/(0.155 - 0.05) = \$21.43$.

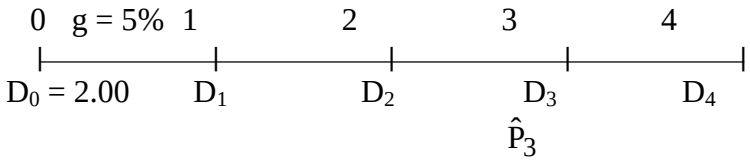
b. $r_s = 9\% + (12\% - 9\%)1.5 = 13.5\%$. $\hat{P}_0 = \$2.25/(0.135 - 0.05) = \$26.47$.

c. $r_s = 9\% + (11\% - 9\%)1.5 = 12.0\%$. $\hat{P}_0 = \$2.25/(0.12 - 0.05) = \$32.14$.

d. New data given: $r_{RF} = 9\%$; $r_M = 11\%$; g = 6%, b = 1.3.

$r_s = r_{RF} + (r_M - r_{RF})b = 9\% + (11\% - 9\%)1.3 = 11.6\%$.
$\hat{P}_0 = D_1/(r_s - g) = \$2.27/(0.116 - 0.06) = \$40.54$.

7-17



a. $D_1 = \$2(1.05) = \$2.10$. $D_2 = \$2(1.05)^2 = \$2.21$. $D_3 = \$2(1.05)^3 = \$2.32$.

b. PV = \$2.10(0.8929) + \$2.21(0.7972) + \$2.32(0.7118) = \$5.29.

Calculator solution: Input 0, 2.10, 2.21, and 2.32 into the cash flow register, input I/YR = 12, PV = ? PV = \$5.29.

c. \$34.73(0.7118) = \$24.72.

Calculator solution: Input 0, 0, 0, and 34.73 into the cash flow register, I/YR = 12, PV = ? PV = \$24.72.

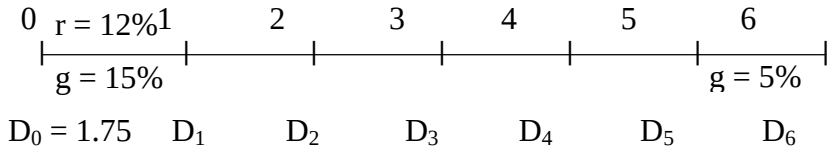
d. \$24.72 + \$5.29 = \$30.01 = Maximum price you should pay for the stock.

e. $$\hat{P}_0 = \frac{D_0(1+g)}{r_s - g} = \frac{D_1}{r_s - g} = \frac{\$2.10}{0.12 - 0.05} = \$30.00.$$

f. The value of the stock is *not* dependent upon the holding period. The value calculated in Parts a through d is the value for a 3-year holding period. It is equal to the value calculated in Part e except for a small rounding error. Any other holding period would produce the same value of $\hat{P}_0$; that is, $\hat{P}_0 = \$30.00$.

© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

7-18 a. End of Year:



$D_t = D_0(1 + g)^t$
$D_1 = \$1.75(1.15)^1 = \$2.01.$
$D_2 = \$1.75(1.15)^2 = \$1.75(1.3225) = \$2.31.$
$D_3 = \$1.75(1.15)^3 = \$1.75(1.5209) = \$2.66.$
$D_4 = \$1.75(1.15)^4 = \$1.75(1.7490) = \$3.06.$
$D_5 = \$1.75(1.15)^5 = \$1.75(2.0114) = \$3.52.$

b. Step 1

PV of dividends = $\sum_{t=1}^{5} \frac{D_t}{(1+r_s)^t}$.

PV $D_1 = \$2.01(PVIF_{12\%,1}) = \$2.01(0.8929) = \$1.79$
PV $D_2 = \$2.31(PVIF_{12\%,2}) = \$2.31(0.7972) = \$1.84$
PV $D_3 = \$2.66(PVIF_{12\%,3}) = \$2.66(0.7118) = \$1.89$
PV $D_4 = \$3.06(PVIF_{12\%,4}) = \$3.06(0.6355) = \$1.94$
PV $D_5 = \$3.52(PVIF_{12\%,5}) = \$3.52(0.5674) = \$2.00$
PV of dividends = \$9.46

Step 2

$$\hat{P}_5 = \frac{D_6}{r_s - g_n} = \frac{D_5(1+g_n)}{r_s - g_n} = \frac{\$3.52(1.05)}{0.12 - 0.05} = \frac{\$3.70}{0.07} = \$52.80.$$

This is the price of the stock 5 years from now. The PV of this price, discounted back 5 years, is as follows:

PV of $\hat{P}_5 = \$52.80(PVIF_{12\%,5}) = \$52.80(0.5674) = \$29.96.$

© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Step 3

The price of the stock today is as follows:

$\hat{P}_0$ = PV dividends Years 1 through 5 + PV of $\hat{P}_5$
= \$9.46 + \$29.96 = \$39.42.

This problem could also be solved by substituting the proper values into the following equation:

$$\hat{P}_0 = \sum_{t=1}^{5} \frac{D_0(1+g_s)^t}{(1+r_s)^t} + \left(\frac{D_6}{r_s - g_n}\right)\left(\frac{1}{1+r_s}\right)^5.$$

Calculator solution: Input 0, 2.01, 2.31, 2.66, 3.06, 56.32 (3.52 + 52.80) into the cash flow register, input I/YR = 12, PV = ? PV = \$39.43.

c. First Year (t = 0)

D_1/P_0 = \$2.01/\$39.42	= 5.10%
Capital gains yield	= 6.90%
Expected total return	= 12.00%

Sixth Year (t = 5)

D_6/P_5 = \$3.70/\$52.80	= 7.00%
Capital gains yield	= 5.00
Expected total return	= 12.00%

*We know that r is 12%, and the dividend yield is 5.10%; therefore, the capital gains yield must be 6.90%.

The main points to note here are as follows:

1. The total yield is always 12% (except for rounding errors).

2. The capital gains yield starts relatively high, then declines as the supernormal growth period approaches its end. The dividend yield rises.

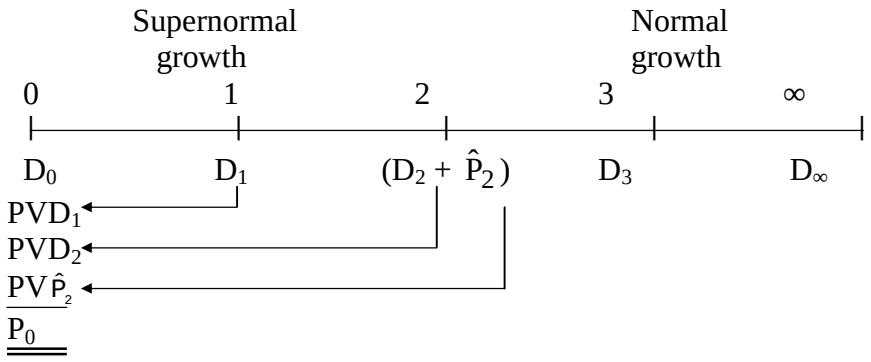
3. After t = 5, the stock will grow at a 5% rate. The dividend yield will equal 7%, the capital gains yield will equal 5%, and the total return will be 12%.

© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

d. If the price as estimated by the marginal investor differs from the market price, then investors will buy or sell until an equilibrium has been established, with the intrinsic value as estimated by the marginal investor equals the actual market price.

If you think the stock is priced above or below its intrinsic value, then you should at least consider buying if the stock is undervalued or selling if it is overvalued. If you turn out to be correct, then you will make money, eventually if you hold on to an unpopular position long enough.

7-19 a. Part 1. Graphical representation of the problem:



$D_1 = D_0(1 + g_s) = \$1.60(1.20) = \$1.92.$
$D_2 = D_0(1 + g_s)^2 = \$1.60(1.20)^2 = \$2.304.$

$$\hat{P}_2 = \frac{D_3}{r_s - g_n} = \frac{D_2(1+g_n)}{r_s - g_n} = \frac{\$2.304\,(1.06)}{0.10 - 0.06} = \$61.06.$$

$$\hat{P}_0 = PV(D_1) + PV(D_2) + PV(\hat{P}_2)$$
$$= \frac{D_1}{(1+r_s)} + \frac{D_2}{(1+r_s)^2} + \frac{\hat{P}_2}{(1+r_s)^2}$$
$$= \$1.92(0.9091) + \$2.304(0.8264) + \$61.06(0.8264) = \$54.11.$$

Calculator solution: Input 0, 1.92, 63.364(2.304 + 61.06) into the cash flow register, input I/YR = 10, PV = ? PV = $54.11.

© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Part 2.

Expected dividend yield: $D_1/P_0 = \$1.92/\$54.11 = 3.55\%$.

Capital gains yield: First, find $\hat{P}_1$ which equals the sum of the present values of D_2 and $\hat{P}_2$, discounted for one year.

$$\hat{P}_1 = D_2/(1.10) + \hat{P}_2/(1.10) = \frac{\$2.304 + \$61.06}{(1.10)^1} = \$57.60.$$

Calculator solution: Input 0, 63.364 (2.304 + 61.06) into the cash flow register, input I/YR = 10, PV = ? PV = $57.60.

Second, find the capital gains yield:

$$\frac{\hat{P}_1 - P_0}{P_0} = \frac{\$57.60 - \$54.11}{\$54.11} = 6.45\%.$$

Dividend yield = 3.55%
Capital gains yield = 6.45
10.00% = r_s.

b. Due to the longer period of supernormal growth, the value of the stock will be higher for each year. Although the total return will remain the same, r_s = 10%, the distribution between dividend yield and capital gains yield will differ: The dividend yield will start off lower and the capital gains yield will start off higher for the 5-year supernormal growth condition, relative to the 2-year supernormal growth state. The dividend yield will increase and the capital gains yield will decline over the 5-year period until dividend yield = 4% and capital gains yield = 6%.

c. Throughout the supernormal growth period, the total yield will be 10%, but the dividend yield is relatively low during the early years of the supernormal growth period and the capital gains yield is relatively high. As we near the end of the supernormal growth period, the capital gains yield declines and the dividend yield rises. After the supernormal growth period has ended, the capital gains yield will equal g_n = 6%. The total yield must equal r_s = 10%, so the dividend yield must equal 10% - 6% = 4%.

d. Some investors need cash dividends (retired people) while others would prefer growth. Also, investors must pay taxes each year on the dividends received during the year, while taxes on capital gains can be delayed until the gain is actually realized.

© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

SOLUTION TO SPREADSHEET PROBLEM

7-20 The detailed solution for the spreadsheet problem, ***Solution for Ch 07 P20 Build a Model.xls***, is available at the textbook's Web site.

© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

MINI CASE

Sam Strother and Shawna Tibbs are senior vice presidents of the Mutual of Seattle. They are co-directors of the company's pension fund management division, with Strother having responsibility for fixed income securities (primarily bonds) and Tibbs being responsible for equity investments. A major new client, the Northwestern Municipal League, has requested that Mutual of Seattle present an investment seminar to the mayors of the cities in the association, and Strother and Tibbs, who will make the actual presentation, have asked you to help them.

To illustrate the common stock valuation process, Strother and Tibbs have asked you to analyze the Temp Force Company, an employment agency that supplies word processor operators and computer programmers to businesses with temporarily heavy workloads. You are to answer the following questions.

a. Describe briefly the legal rights and privileges of common stockholders.

Answer: The common stockholders are the owners of a corporation, and as such, they have certain rights and privileges as described below.

1. Ownership implies control. Thus, a firm's common stockholders have the right to elect its firm's directors, who in turn elect the officers who manage the business.

2. Common stockholders often have the right, called the preemptive right, to purchase any additional shares sold by the firm. In some states, the preemptive right is automatically included in every corporate charter; in others, it is necessary to insert it specifically into the charter.

© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

b. 1. Write out a formula that can be used to value any stock, regardless of its dividend pattern.

Answer: The value of any stock is the present value of its expected dividend stream:

$$\hat{P}_0 = \frac{D_1}{(1+r_s)^t} + \frac{D_2}{(1+r_s)} + \frac{D_3}{(1+r_s)^3} + \cdots + \frac{D_\infty}{(1+r_s)^\infty}.$$

However, some stocks have dividend growth patterns which allow them to be valued using short-cut formulas.

b. 2. What is a constant growth stock? How are constant growth stocks valued?

Answer: A constant growth stock is one whose dividends are expected to grow at a constant rate forever. "Constant growth" means that the best estimate of the future growth rate is some constant number, not that we really expect growth to be the same each and every year. Many companies have dividends which are expected to grow steadily into the foreseeable future, and such companies are valued as constant growth stocks.

For a constant growth stock:

$$D_1 = D_0(1 + g),\ D_2 = D_1(1 + g) = D_0(1 + g)^2, \text{ and so on.}$$

With this regular dividend pattern, the general stock valuation model can be simplified to the following very important equation:

$$\hat{P}_0 = \frac{D_1}{r_s - g} = \frac{D_0(1+g)}{r_s - g}.$$

This is the well-known "Gordon," or "constant-growth" model for valuing stocks. Here D_1, is the next expected dividend, which is assumed to be paid 1 year from now, r_s is the required rate of return on the stock, and g is the constant growth rate.

b. 3. What happens if a company has a constant g that exceeds its r_s? Will many stocks have expected $g > r_s$ in the short run (i.e., for the next few years)? In the long run (i.e., forever)?

© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Answer: The model is derived mathematically, and the derivation requires that $r_s > g$. If g is greater than r_s, the model gives a negative stock price, which is nonsensical. The model simply cannot be used unless (1) $r_s > g$, (2) g is expected to be constant, and (3) g can reasonably be expected to continue indefinitely.

Stocks may have periods of supernormal growth, where $g_s > r_s$; however, this growth rate cannot be sustained indefinitely. In the long-run, $g < r_s$.

© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

c. **Assume that Temp Force has a beta coefficient of 1.2, that the risk-free rate (the yield on T-bonds) is 7%, and that the market risk premium is 5%. What is the required rate of return on the firm's stock?**

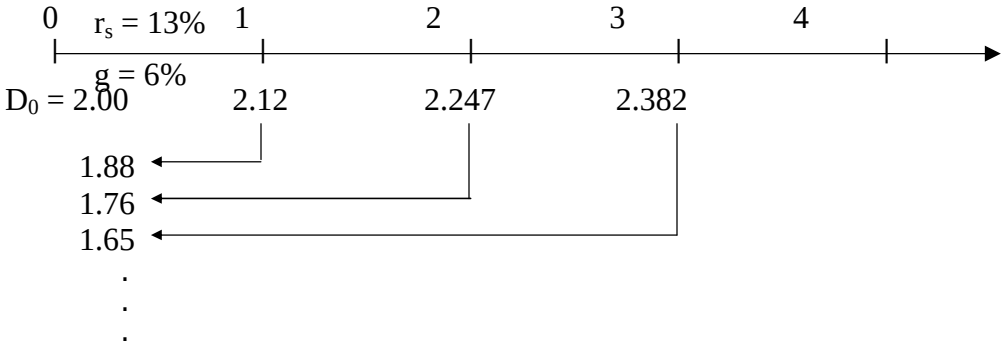
Answer: Here we use the SML to calculate temp force's required rate of return:

$$r_s = r_{RF} + (r_M - r_{RF})b_{Temp\ Force} = 7\% + (12\% - 7\%)(1.2)$$
$$= 7\% + (5\%)(1.2) = 7\% + 6\% = 13\%.$$

d. **Assume that Temp Force is a constant growth company whose last dividend (D_0, which was paid yesterday) was $2.00 and whose dividend is expected to grow indefinitely at a 6% rate.**

d. 1. What is the firm's expected dividend stream over the next 3 years?

Answer: Temp Force is a constant growth stock, and its dividend is expected to grow at a constant rate of 6% per year. Expressed as a time line, we have the following setup. Just enter 2 in your calculator; then keep multiplying by 1 + g = 1.06 to get D_1, D_2, and D_3:



© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

d. 2. What is the firm's current stock price?

Answer: We could extend the time line on out forever, find the value of Temp Force's dividends for every year on out into the future, and then the PV of each dividend, discounted at r = 13%. For example, the PV of D_1 is \$1.76106; the PV of D_2 is \$1.75973; and so forth. Note that the dividend payments increase with time, but as long as $r_s > g$, the present values decrease with time. If we extended the graph on out forever and then summed the PVs of the dividends, we would have the value of the stock. However, since the stock is growing at a constant rate, its value can be estimated using the constant growth model:

$$\hat{P}_0 = \frac{D_1}{r_s - g} = \frac{\$2.12}{0.13 - 0.06} = \frac{\$2.12}{0.07} = \$30.29.$$

d. 3. What is the stock's expected value one year from now?

Answer: After one year, D_1 will have been paid, so the expected dividend stream will then be D_2, D_3, D_4, and so on. Thus, the expected value one year from now is \$32.10:

$$\hat{P}_1 = \frac{D_2}{(r_s - g)} = \frac{\$2.247}{(0.13 - 0.06)} = \frac{\$2.247}{0.07} = \$32.10.$$

© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

d. 4. What are the expected dividend yield, the capital gains yield, and the total return during the first year?

Answer: The expected dividend yield in any year n is

$$\text{Dividend Yield} = \frac{D_n}{\hat{P}_{n-1}},$$

While the expected capital gains yield is

$$\text{Capital Gains Yield} = \frac{(\hat{P}_n - \hat{P}_{n-1})}{\hat{P}_{n-1}} = r - \frac{D_n}{P_{n-1}}.$$

Thus, the dividend yield in the first year is 10%, while the capital gains yield is 6%:

Total return	= 13.0%
Dividend yield = \$2.12/\$30.29	= 7.0%
Capital gains yield	= 6.0%

e. Now assume that the stock is currently selling at \$30.29. What is its expected rate of return?

Answer: The constant growth model can be rearranged to this form:

$$\hat{r}_s = \frac{D_1}{P_0} + g.$$

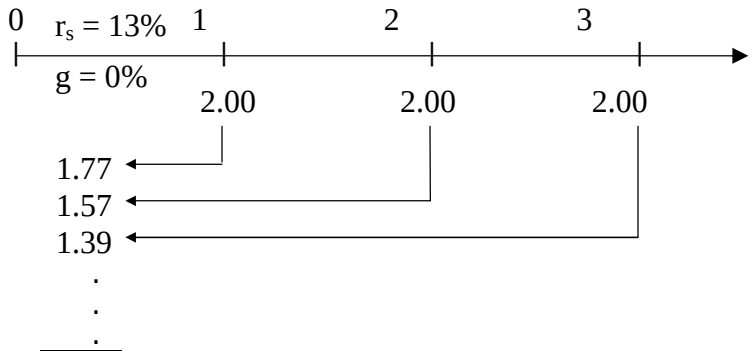
Here the current price of the stock is known, and we solve for the expected return. For Temp Force:

$$\hat{r}_s = \$2.12/\$30.29 + 0.060 = 0.070 + 0.060 = 13\%.$$

© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

f.	**What would the stock price be if its dividends were expected to have zero growth?**

Answer: If Temp Force's dividends were not expected to grow at all, then its dividend stream would be a perpetuity. Perpetuities are valued as shown below:



P_0 = 15.38

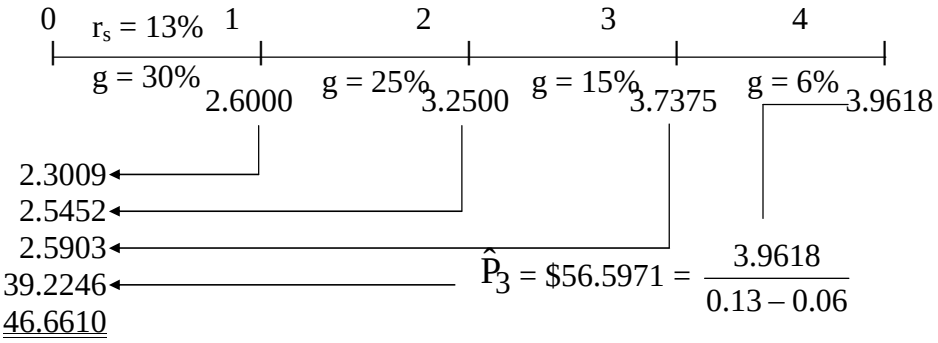
P_0 = PMT/r = \$2.00/0.13 = \$15.38.

Note that if a preferred stock is a perpetuity, it may be valued with this formula.

© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

g. **Now assume that Temp Force's dividend is expected to experience supernormal growth of 30% from Year 0 to Year 1, 20% from Year 1 to Year 2, and 10% from Year 2 to Year 3. After Year 3, dividends will grow at a constant rate of 6%. What is the stock's intrinsic value under these conditions? What are the expected dividend yield and capital gains yield during the first year? What are the expected dividend yield and capital gains yield during the fourth year (from Year 3 to Year 4)?**

Answer: Temp Force is no longer a constant growth stock, so the constant growth model is not applicable. Note, however, that the stock is expected to become a constant growth stock in 3 years. Thus, it has a nonconstant growth period followed by constant growth. The easiest way to value such nonconstant growth stocks is to set the situation up on a time line as shown below:



Simply enter \$2 and multiply by (1.30) to get D_1 = \$2.60; multiply that result by 1.25 to get D_2 = \$3.25, multiply that result by 1.15 to get D_3 = \$3.7375 and multiply that result by 1.06 to get D_4 = \$3.9618. Then recognize that after year 3, Temp Force becomes a constant growth stock, and at that point $\hat{P}_3$ can be found using the constant growth model. $\hat{P}_3$ is the present value as of t = 3 of the dividends in year 4 and beyond.

With the cash flows for D_1, D_2, D_3, and $\hat{P}_3$ shown on the time line, we discount each value back to year 0, and the sum of these four PVs is the intrinsic value of the stock today, $\hat{P}_0$ = \$46.6610 ≈ \$46.66.

© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

The dividend yield and the capital gains yield are:

$$\text{Dividend yield} = \frac{\$2.600}{\$46.66} = 0.0557 \approx 5.6\%.$$

$$\text{Capital gains yield} = 13.00\% - 5.6\% = 7.4\%.$$

During the nonconstant growth period, the dividend yields and capital gains yields are not constant, and the capital gains yield does not equal g. However, after year 3, the stock becomes a constant growth stock, with g = capital gains yield = 6.0% and dividend yield = 13.0% - 6.0% = 7.0%.

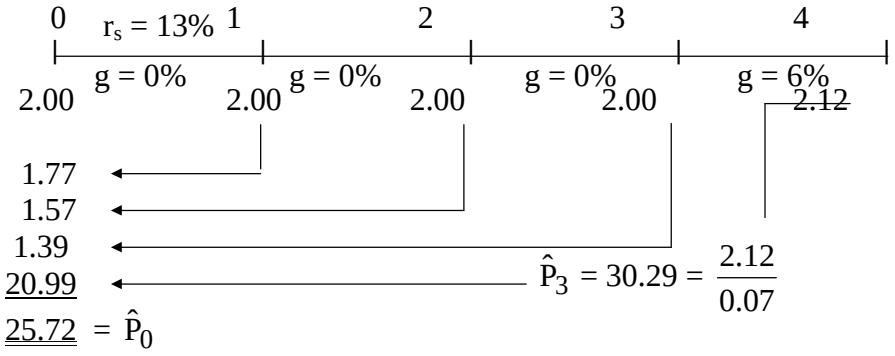
h.	**Is the stock price based more on long-term or short-term expectations? Answer this by finding the percentage of Temp Force's current stock price based on dividends expected more than three years in the future.**

Answer: $\frac{\$39.2246}{\$46.6610} = 84.1\%.$

Stock price is based more on long-term expectations, as is evident by the fact that almost 85% of temp force stock price is determined by dividends expected more than three years from now.

i.	**Suppose Temp Force is expected to experience zero growth during the first 3 years and then to resume its steady-state growth of 6% in the fourth year. What is the stock's value now? What is its expected dividend yield and its capital gains yield in Year 1? In Year 4?**

Answer: Now we have this situation:



© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

During year 1:

$$\text{Dividend Yield} = \frac{\$2.00}{\$25.72} = 0.0778 = 7.78\%.$$

$$\text{Capital Gains Yield} = 13.00\% - 7.78\% = 5.22\%.$$

Again, in year 4 Temp Force becomes a constant growth stock; hence g = capital gains yield = 6.0% and dividend yield = 7.0%.

j. Finally, assume that Temp Force's earnings and dividends are expected to decline by a constant 6% per year forever, that is, g = −6%. Why would anyone be willing to buy such a stock, and at what price should it sell? What would be the dividend yield and capital gains yield in each year?

Answer: The company is earning something and paying some dividends, so it clearly has a value greater than zero. That value can be found with the constant growth formula, but where g is negative:

$$P_0 = \frac{D_1}{r_S - g} = \frac{D_0(1+g)}{r_S - g} = \frac{\$2.00(0.94)}{0.13 - (-0.06)} = \frac{\$1.88}{0.19} = \$9.89.$$

Since it is a constant growth stock:

$$g = \text{Capital Gains Yield} = -6.0\%,$$

hence:

$$\text{Dividend Yield} = 13.0\% - (-6.0\%) = 19.0\%.$$

As a check:

$$\text{Dividend Yield} = \frac{\$1.88}{\$9.89} = 0.190 = 19.0\%.$$

The dividend and capital gains yields are constant over time, but a high (19.0%) dividend yield is needed to offset the negative capital gains yield.

© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

k. What is market mutliple analysis?

Answer: Analysts often use the P/E multiple (the price per share divided by the earnings per share) or the P/CF multiple (price per share divided by cash flow per share, which is the earnings per share plus the dividends per share) to value stocks. For example, estimate the average P/E ratio of comparable firms. This is the P/E multiple. Multiply this average P/E ratio by the expected earnings of the company to estimate its stock price. The entity value (V) is the market value of equity (# shares of stock multiplied by the price per share) plus the value of debt. Pick a measure, such as EBITDA, sales, customers, eyeballs, etc. Calculate the average entity ratio for a sample of comparable firms. For example, V/EBITDA, V/customers. Then find the entity value of the firm in question. For example, multiply the firm's sales by the V/sales multiple, or multiply the firm's # of customers by the V/customers ratio. The result is the total value of the firm. Subtract the firm's debt to get the total value of equity. Divide by the number of shares to get the price per share. There are problems with market multiple analysis. (1) It is often hard to find comparable firms. (2) The average ratio for the sample of comparable firms often has a wide range. For example, the average P/E ratio might be 20, but the range could be from 10 to 50. How do you know whether your firm should be compared to the low, average, or high performers?

l. Temp Force recently issued preferred stock that pays an annual dividend of $5 at a price of $50 per share. What is the expected return to an investor who buys this preferred stock?

Answer:

$$\hat{r}_{ps} = \frac{D_{ps}}{V_{ps}}$$

$$= \frac{\$5}{\$50}$$

$$= 10\%.$$

© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

m. **Why do stock prices change? Suppose the expected D_1 is \$2, the growth rate is 5%, and r_s is 10%. Using the constant growth model, what is the stock's price? What is the impact on the stock price if g falls to 4% or rises to 6%? If r_s increases to 9% or to 11%?**

Answer: Using the constant growth model, the price of a stock is $P_0 = D_1 / (r_s - g)$. If estimates of g change, then the price will change. If estimates of the required return on stock change, then the stock price will change. Notice that $r_S = r_{RF} + (RP_M)b_i$, so r_s will change if there are changes in inflation expectations, risk aversion, or company risk. The following table shows the stock price for various levels of g and r_s.

	g	g	g
r_S	4%	5%	6%
9%	40.00	50.00	66.67
10%	33.33	40.00	50.00
11%	28.57	33.33	40.00

n. **What does market equilibrium mean?**

Answer: Equilibrium means stable, no tendency to change. Market equilibrium means that prices are stable--at its current price, there is no general tendency for people to want to buy or to sell a security that is in equilibrium. Also, when equilibrium exists, the expected rate of return will be equal to the required rate of return:

$$\hat{r} = D_1/P_0 + g = r = r_{RF} + (r_M - r_{RF})b.$$

o. **If equilibrium does not exist, how will it be established?**

Answer: Securities will be bought and sold until the equilibrium price is established.

© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

p. **What is the Efficient Markets Yypothesis, what are its three forms, and what are its implications?**

Answer: The EMH in general is the hypothesis that securities are normally in equilibrium, and are "priced fairly," making it impossible to "beat the market."

Weak-form efficiency says that investors cannot profit from looking at past movements in stock prices--the fact that stocks went down for the last few days is no reason to think that they will go up (or down) in the future. This form has been proven pretty well by empirical tests, even though people still employ "technical analysis."

Semistrong-form efficiency says that all publicly available information is reflected in stock prices, hence that it won't do much good to pore over annual reports trying to find undervalued stocks. This one is (I think) largely true, but superior analysts can still obtain and process new information fast enough to gain a small advantage.

Strong-form efficiency says that all information, even inside information, is embedded in stock prices. This form does not hold--insiders know more, and could take advantage of that information to make abnormal profits in the markets. Trading on the basis of insider information is illegal.

q. **Assume that all the growth rates used in the preceding answers were averages of the growth rates published by well known and respected security analysts. This being the case, would you say that your results are based on a purely rational analysis? If not, what factors might have led to "irrational results?"**

Answer: Security analysts are people too, and they can be subject to animal spirits, herding instincts, and anchoring biases like the rest of us. So, the results might de facto be irrational even though those involved were trying hard to behave rationally.

This is different from the person who thinks stocks are way overpriced but buys anyway, thinking he or she can find an even bigger fool to sell the stock to at a profit. This could be sort of a rational exploitation of irrational behavior.

We must say, though, that it's a lot easier to make good decisions with hindsight than with only foresight.

© 2011 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.